

The signed-reflection identity for arbitrary continuous integrands

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Abstract

Unit 216 (Astra #23 programme D2, step 1). Unit 196 reduced the zero-phase dressed integral over the symmetric box $(-1, 1]^d$ to the unit box by reflecting coordinates; with a phase the kernel is no longer a function of x^{2k} alone, so the reduction is needed for an arbitrary continuous integrand:

$$\int_{(-1, 1]^d} F(x) dx = \sum_{\sigma \in \{\pm 1\}^d} \int_{(0, 1]^d} F(\sigma \cdot u) du \quad (\text{integral_symBox_eq_sum_reflect}).$$

Proof by induction on d with the Bochner box peel, the one-coordinate symmetrisation $\int_{-1}^1 g = \int_0^1 (g(a) + g(-a))$, and the splitting of the sign sum along the first coordinate (`Fin.consEquiv`).
Zero sorry/axiom; 9093 jobs.

1 The things formalised

```
theorem integrableOn_piBox_Ioc_of_continuous (d) (a b : ) (F) (hF : Continuous F) :
  IntegrableOn F (piBox d (Ioc a b))
theorem integral_symBox_eq_sum_reflect : (d : ) (F : (Fin d → ) → ), Continuous F →
  x in symBox d, F x = : Fin d → Bool, u in unitBox d, F (reflect u)
```

2 Mathematics

For $d + 1$ coordinates: peel the first, $\int_{-1}^1 \int_{(-1, 1]^d} F(a, b) db da$; the inner integral is $\sum_{\tau} \int_{(0, 1]^d} F(a, \tau \cdot v) dv$ by induction; symmetrise a : $\int_0^1 \sum_{\tau} [\int F(a, \tau v) + \int F(-a, \tau v)] da$; on the right, $\sum_{\sigma=(b, \tau)} \int_{(0, 1]^{d+1}} F(\sigma \cdot u)$ peels to $\sum_b \sum_{\tau} \int_0^1 \int F(\text{sgn}(b)a, \tau v)$ by `reflect_cons`; the two agree with $\text{sgn}(\text{true}) = -1$, $\text{sgn}(\text{false}) = 1$. The marginal integrabilities come from `integrableOn_pi_box_marginal` applied to $F \circ \text{reflect}$.

3 Lean notes

- `reflect`, `sgn`, `integral_Ioc_symm_add` live in `SignedReflection.lean`, a sibling branch of the import graph not reachable from the phase programme; import `SymmetricAmplitudeAsymptotic` explicitly.
- In `integral_pi_box_succ d _ _ (proof)` the integrand gets unified from the proof term `(F reflect , a Function.comp)` rather than from the goal's lambda, and the rewrite then fails; pass the lambda explicitly as the function argument.

- `simp only [sgn, Bool.false_eq_true, ↓reduceIte, neg_one_mul, one_mul]` evaluates `sgn true * a` and `sgn false * a` without disturbing the integrals.

4 Next

Unit 217: Headline XIX — the phase-dressed leading term on the symmetric box as the sum over orthants of Headline XIII with phase $\varepsilon_\sigma \xi \circ \sigma$ ($\varepsilon_\sigma = \prod \text{sgn}(\sigma_i)^{k_i}$) and amplitude $\chi_h(\sigma) \eta \circ \sigma$ (signed monomial) or $\eta \circ \sigma$ (absolute density $|x|^h$).